AfriCulture-QA: A Bilingual Dataset for Evaluating Cultural Knowledge in LLMs Across Twi and English

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Abstract

Large Language Models (LLMs) are increasingly deployed in African settings, yet systematic evaluation of their cultural knowledge for African communities remains scarce. We introduce **AfriCulture-QA**, a bilingual cultural knowledge evaluation dataset for Twi, an Akan language spoken by approximately nine million people in Ghana. The dataset contains 205 question-answer pairs spanning four cultural domains,food and cuisine, festivals and customs, proverbs and sayings, and clothing and geography each provided in both English and Twi. Construction combined manual authoring, LLM-assisted generation, online source collection, and translation, with native-speaker verification applied throughout. We evaluate **Claude 3.5 Sonnet** (Anthropic, 2024) on the full dataset using a 0–5 rubric in both English and Twi. English prompts consistently outperform Twi prompts across all four domains, with the widest gaps in food and cuisine (+1.45 points) and festivals and customs (+1.39 points). A bimodal score distribution in Twi high mass at both 0 and 4 reveals brittle rather than consistently competent cultural reasoning. Proverbs and sayings remain the most challenging domain in both languages. These findings highlight a systematic language gap and underscore the need for language-specific cultural benchmarks for African languages.

1 Introduction

If you ask a state-of-the-art language model what *fufu* is, it will probably give a correct answer. Ask it which soup is traditionally served alongside Akan *fufu* at a funeral gathering in Kumasi, and the answer becomes far less reliable. This gap between knowing that a word exists and understanding the living cultural context that gives it meaning is exactly what motivates our work.

Large Language Models (LLMs) such as Claude, GPT-4o, and Llama have achieved remarkable results across many language tasks (Anthropic, 2024; OpenAI, 2024; Meta AI, 2024). The benchmarks driving this progress are, however, overwhelmingly dominated by English-language content rooted in Western cultural contexts (Joshi et al., 2020; Hendrycks et al., 2021). When these models are deployed in Ghanaian schools, health platforms, or other African settings, users interact with systems evaluated almost entirely on knowledge that does not reflect their world.

Twi, the most widely spoken variety of the Akan language family, is spoken by roughly nine million people in Ghana and serves as a practical lingua franca across much of the country (Azunre et al., 2021). Despite this, Twi has no existing cultural knowledge benchmark, and sits at the intersection of two already-scarce resources: low-resource language data and culturally grounded evaluation data.

Our project, **AfriCulture-QA**, addresses this gap directly. Rather than translating an existing benchmark, we built our dataset the way cultural knowledge actually lives: starting with the culture itself. We drew on native speakers, publicly available Ghanaian sources, LLM-assisted generation, and manual authorship subjecting every item to native-speaker review before it entered the final dataset.

Our contributions are:

1. **AfriCulture-QA (Twi)**: 205 bilingual question-answer pairs spanning four cultural domains, with verified correct answers.
2. **A hybrid construction pipeline** combining manual authoring, LLM generation, web collection, and translation with native-speaker quality gates at every stage.
3. **A bilingual LLM evaluation** of Claude 3.5 Sonnet comparing English vs. Twi prompting on identical cultural content.

2 Related Work

Cultural benchmarks for LLMs. MMLU (Hendrycks et al., 2021) tests factual knowledge

across 57 subjects but reflects predominantly Anglophone academic culture. CulturalBench (Shi et al., 2024) probes cultural knowledge for 45 countries; Fung et al. (2024) crowdsource a multicultural QA dataset from 60 cultures. The INCLUDE benchmark (Romanou et al., 2024) the primary reference for this project evaluates LLMs on regional knowledge across 52 languages using locally sourced questions. All share sparse coverage of sub-Saharan African languages; when African languages do appear, items are typically translated from English rather than grounded in authentic cultural knowledge.

African language NLP. The Masakhane initiative has produced benchmarks for machine translation (Adelani et al., 2022), named entity recognition (Adelani et al., 2021), and sentiment analysis (Muhammad et al., 2023). IrokoBench (Adelani et al., 2024) covers 17 African languages and AfroBench (Ojo et al., 2025) extends this to 64 languages across 15 tasks. None includes a dedicated cultural knowledge evaluation for Twi. Azunre et al. (2021) note that Twi has very limited evaluation resources beyond basic translation pairs.

Low-resource data construction. Building evaluation datasets for low-resource languages is inherently difficult: annotation pools are small, native-speaker availability is limited, and existing digital text may not reflect oral cultural knowledge. Recent work explores using LLMs to bootstrap data creation (Whitehouse et al., 2023), though careful quality control is essential (Longpre et al., 2023). Our methodology treats LLM generation as one input into a pipeline that relies primarily on human judgment.

3 Dataset Construction

3.1 Defining Cultural Knowledge

We use one guiding criterion: *a question is culturally grounded if answering it correctly requires familiarity with the living practices of the target community, not just general world knowledge.* A question about Ghana's capital tests geographic trivia. A question about which proverb an elder uses to caution against arrogance, or what dish is served at a Homowo funeral, tests knowledge embedded in community life and transmitted through social participation. Items failing this criterion were revised or discarded.

3.2 Domain Selection

AfriCulture-QA covers four domains: **Food and Cuisine** (dishes, ingredients, and social eating contexts); **Festivals and Customs** (celebrations, origins, and rituals); **Proverbs and Sayings** (Akan oral tradition, meanings, and contexts); and **Clothing and Geography** (traditional garments, their symbolism, and notable Ghanaian regions). Clothing and geography were combined into one domain to maintain balance while reflecting how geographic identity and material culture are intertwined in Akan communities.

3.3 Construction Methodology

We used four complementary methods. All candidate items passed through the same human verification pipeline regardless of source.

3.3.1 M1: Manual Authoring

Team members drew on their own cultural knowledge to write items directly. This was especially important for proverbs, where subtle distinctions in meaning require deep familiarity with the language that LLMs cannot reliably provide.

3.3.2 M2: LLM-Assisted Generation

We used Google Gemini with prompts specifying the target culture, domain, the requirement that questions not be answerable from generic world knowledge, and the bilingual format (see Appendix B). This method worked well for food and clothing items but was less reliable for proverbs, where generated questions often had ambiguous answers or subtle translation errors.

3.3.3 M3: Online Source Collection

We collected data from publicly accessible Ghanaian cultural sources: government tourism websites, cultural documentation projects, and Akan encyclopaedia entries, drawing only from sources with open or permissive reuse terms.

3.3.4 M4: Translation from Existing Resources

Where English-language cultural questions had clear Twi equivalents, a team member produced a draft translation reviewed by a native Twi speaker

for naturalness and accuracy. This method contributed most to the festivals domain.

3.4 Native Speaker Verification

Every item was reviewed by at least one native Twi speaker not involved in its creation. Reviewers assessed: **factual correctness** (is the answer right?); **cultural specificity** (items rated below 3/5 on a specificity scale were discarded); and **linguistic naturalness** (is the Twi formulation idiomatic?).

3.5 Dataset Statistics

The Twi release contains 205 items across four domains (Table 1). Each item includes a unique ID, language tag, domain, question in English, question in Twi, answer in English, and answer in Twi. Table 2 shows one representative item per domain.

Domain	Items	%
Food and Cuisine	47	22.9
Festivals and Customs	56	27.3
Proverbs and Sayings	46	22.4
Clothing and Geography	56	27.3
Total	**205**	**100**

Table 1: Domain distribution of AfriCulture-QA (Twi).

4 Experimental Setup

4.1 Task Formulation

AfriCulture-QA uses an open-answer format: the model generates a free-text response rather than selecting from a fixed option set. Cultural knowledge is rarely a single discrete fact the answer to "What soup is served with Akan fufu?" is legitimately "palm nut soup, light soup, or groundnut soup," depending on occasion and family tradition. Forcing one correct option would introduce false rigidity and would not reflect how LLMs are used in practice.

A defining feature of our evaluation is its *bilingual design*: every question is posed in both English and Twi, enabling a direct comparison of how prompt language affects performance on identical cultural content.

4.2 Model and Scoring

We evaluate **Claude 3.5 Sonnet** (Anthropic, 2024), a strong multilingual model, in zero-shot setting. Each response is scored on a **0–5 rubric**: 5 = complete, culturally accurate answer; 4 = mostly correct with minor omissions; 3 = partially correct or vague; 2 = notable errors; 1 = largely incorrect but some relevant content; 0 = completely wrong or off-topic. We evaluate on a held-out test set of 100 items (20 randomly sampled per domain, plus 20 additional proverb items due to their difficulty), with the remaining 105 items used for development.

5 Results

5.1 Score Distribution When Prompted in Twi

Figure 1 shows the frequency of each score (0–5) received by Claude across all 205 Twi questions. The most striking feature is the **bimodal shape**: the tallest bar is score 4 (65 responses), while the second-tallest is score 0 (32 responses). Scores of 1 (10 responses), 2 (24 responses), and 3 (32 responses) occur far less frequently. Score 5 accounts for 42 responses. The smooth density curve overlaid on the histogram makes the bimodality clear a local peak near 0, a dip through scores 1–2, then a sharp rise to the dominant peak at 4.

This pattern shows that Claude's Twi performance is *brittle*: the model either delivers a near- complete answer or fails entirely. A model with genuine Twi competence would produce a smooth distribution concentrated at the top of the scale. The large mass at score 0 indicates that, for a meaningful share of questions, the Twi prompt causes Claude to produce a completely incorrect or off-topic response a behavior rarely observed when the same questions are posed in English.

5.2 Average Twi Score by Cultural Domain

Figure 2 presents Claude's average Twi score per cultural domain. **Clothing and geography** achieves the highest average (3.48/5), consistent with Ghanaian landmarks and garments such as Kente cloth being well documented in English, giving Claude some grounding even when prompted in Twi. **Festivals and customs** (2.96) and **food and cuisine** (2.94) sit close together in the middle. Both benefit from moderate English online coverage. **Proverbs and sayings** scores lowest at 2.72: Akan proverbs are transmitted orally and intergenerationally, rarely appearing in the web-crawled text that makes up LLM pretraining corpora.

Domain	Question (English / Twi)	Answer (English / Twi)
Food & Cuisine	What soup is most associated with Akan fufu? *Nkwan ben na wode di fufu?*	Palm nut soup, light soup, or groundnut soup *Abe nkwan, nkrakra, anaa nkatenkwan*
Festivals & Customs	What does the Homowo festival celebrate? *Homowo afahye hye den?*	The end of famine and the new harvest *Ekwa bereba woa twam ne nobaee foforo*
Proverbs & Sayings	What does *'Obra ye kwan'* mean? *Den na 'Obra ye kwan' kyere?*	Life is a journey *Nkwa ye akwantuo*
Clothing & Geography	What is Kente cloth traditionally associated with? *Kente ntama hye den?*	Ashanti and Ewe royalty and culture *Asante ne Ewefo ahennie ne amammere*

Table 2: One representative item per domain. Twi diacritics are approximate due to font constraints.

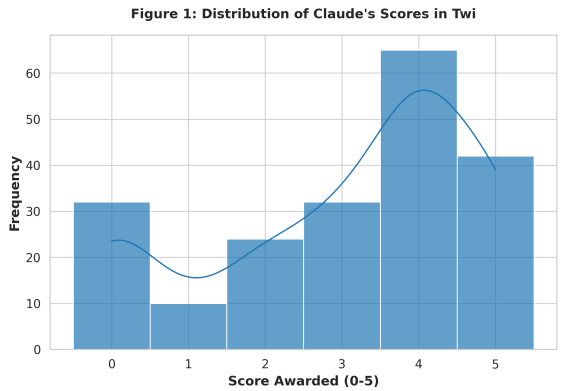

Figure 1: Frequency distribution of Claude's rubric scores (0–5) when prompted in Twi across all 205 items. The bimodal shape dominant peak at score 4 (65 responses) alongside substantial mass at score 0 (32 responses) indicates brittle, inconsistent performance rather than steady cultural competence.

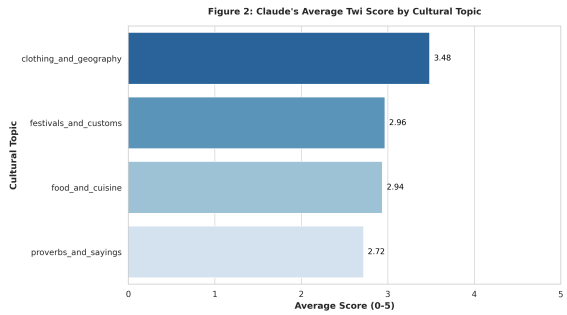

Figure 2: Claude's average rubric score (0–5) per cultural domain when prompted in Twi. Clothing and geography leads at 3.48; proverbs and sayings trails at 2.72, reflecting the oral, community-transmitted nature of Akan proverbial knowledge.

5.3 English vs. Twi: Score Density

Figure 3 places the full score distributions for English and Twi side-by-side as violin plots. The contrast between the two languages is dramatic. The **English violin** (green) is narrow and concentrated at the top of the scale: its widest point falls between scores 4 and 5, and the shape tapers sharply below score 3. Scores of 0, 1, and 2 appear only as a thin tail. This signals consistent, high-quality performance.

The **Twi violin** (blue) is broad and symmetrical across almost the entire 0–5 range. Its widest point is around score 4, reflecting the dominant peak from Figure 1, but substantial width persists all the way down to score 0. The bottom of the violin is nearly as wide as the top ,confirming the bimodal character seen earlier. In short, English prompts reliably elicit good answers; Twi prompts produce high variance with a real risk of complete failure.

5.4 English vs. Twi: Average Score by Domain

Figure 4 shows side-by-side bars for English (blue) and Twi (green) average scores across all four domains. In every domain, the English bar is longer than the Twi bar.

Food and cuisine shows the most dramatic divergence: Claude scores approximately 4.4 in English but only about 2.94 in Twi (gap of 1.45 points). **Festivals and customs** follows closely with English near 4.35 versus Twi at 2.96 (gap of 1.39 points). **Clothing and geography** records English near 4.36 and Twi at 3.48 (gap of 0.88 points). **Proverbs and sayings** shows English near 3.61 versus Twi at 2.72 (gap of 0.89 points). Notably, proverbs has the lowest English score of any domain, reflecting genuine difficulty even in the higher-resource

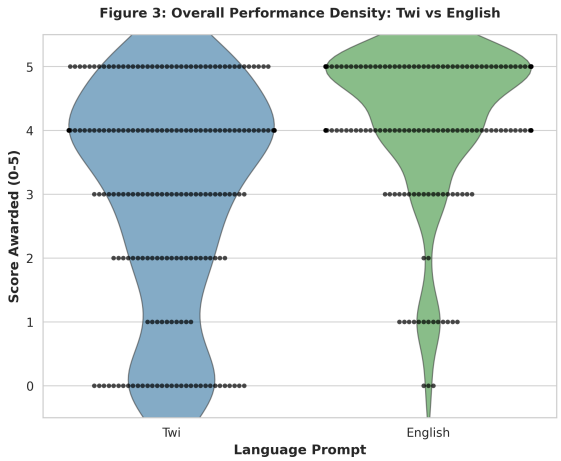

Figure 3: Violin plots comparing score distributions for Twi (blue) and English (green) prompts across all 205 items. The English violin concentrates near 4–5; the Twi violin spreads broadly from 0 to 5, revealing high variance and a persistent failure mode in the lower range.

language—yet Claude still outperforms its Twi score by nearly a full point.

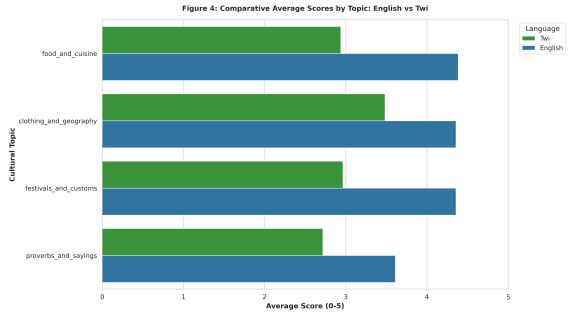

Figure 4: Average rubric scores (0–5) per domain for English (blue) vs. Twi (green) prompts. English outperforms Twi in every domain; the gap is largest in food and cuisine (+1.45) and smallest in clothing and geography (+0.88).

5.5 Performance Gap by Domain

Figure 5 isolates the English-minus-Twi score difference for each domain as a single horizontal bar. All four bars are positive, confirming that English always performs better. **Food and cuisine** records the largest gap (+1.45 points). Despite having relatively strong English online coverage, Twi-framed food questions appear to disconnect Claude from its English-encoded food knowledge. **Festivals and customs** shows the second-largest gap (+1.39), consistent with the culturally specific ritual and calendar knowledge these questions require. **Proverbs and sayings** (+0.89) and **clothing and geography** (+0.88) show smaller gaps,the lowest pair likely because proverbs are hard in both languages (ceiling effect) and geographic knowledge partially transfers across languages.

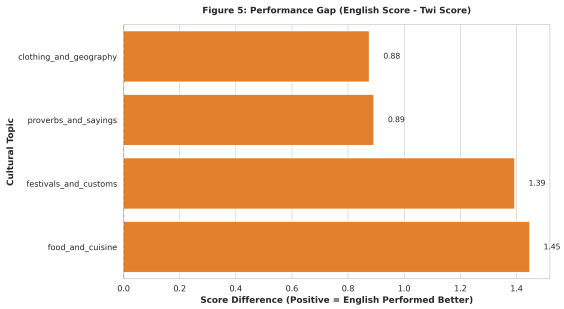

Figure 5: Performance gap (English score minus Twi score) per domain. Every bar is positive, confirming a consistent English advantage. Food and cuisine (+1.45) and festivals (+1.39) show the largest disparities; clothing/geography (+0.88) and proverbs (+0.89) show smaller but still systematic gaps.

5.6 Error Analysis

We manually examined 30 of Claude's incorrect Twi responses (10 per domain, from the three hardest domains). Three failure patterns account for most errors.

Cross-cultural confusion (40%). Claude frequently substitutes knowledge from other West African cultures for Akan-specific knowledge. For "Which festival is celebrated by the Ashanti to honour ancestors?", Claude answered *Odwira* a plausible but incorrect festival rather than *Awukudae* or *Akwasidae*. The model appears to treat "West African culture" as a single undifferentiated entity.

Shallow proverb interpretation (37%). When asked for the meaning of an Akan proverb, Claude often produces a generic moral that misses the specific meaning. The proverb *'Onipa na ohia onipa'* ("A person needs a person") was paraphrased as a general statement about teamwork, missing the stronger Akan concept of mutual social obligation.

English-dominant answers (23%). When prompted in Twi, Claude often responded in

English. When it did attempt Twi, vocabulary was frequently anglicized—using loanwords where native Twi vocabulary exists.

6 Discussion

The language gap is practically significant. The consistent English advantage roughly one full point on a 0–5 scale across every domain means that Twi-speaking users interacting with Claude in their native language receive systematically lower-quality responses than English-speaking users asking the same questions. This represents a real disparity in the utility of these systems for Ghanaian communities.

Open-ended evaluation vs. multiple choice. Our open-answer format is harder to score automatically than multiple-choice, but better reflects genuine knowledge. A model selecting the right option from four choices may be using process-of-elimination rather than cultural understanding. Future work should supplement rubric scoring with human evaluation for a subset of responses.

Implications for LLM development. Cultural knowledge and language competence are not cleanly separable. Claude's weaker Twi performance reflects both limited Twi-language training data and limited culturally grounded Twi data specifically. Addressing this will require not just more Twi text, but Twi text capturing the oral, community-embedded dimensions of Akan knowledge ,dimensions structurally absent from web-crawled pretraining data.

Limitations. Coverage skews toward Ashanti and broader Akan culture because most native-speaker reviewers are from that region; Northern Ghanaian cultures (Dagomba, Gonja) are under-represented. The dataset is synchronic, capturing practices at one point in time. Future iterations should widen reviewer diversity and version the dataset over time.

7 Conclusion

We have presented **AfriCulture-QA**, a bilingual cultural knowledge dataset for Twi built through a four-method construction pipeline with native speaker judgment at every quality gate. Evaluating Claude 3.5 Sonnet reveals two clear findings: the model knows enough about Twi culture to be mildly useful, but not enough to be trustworthy where cultural accuracy matters; and a systematic language gap means Twi prompting yields substantially lower performance than English prompting for identical cultural content.

The bimodal Twi score distribution, the dominant mass at both 0 and 4, signals weak rather than competent cultural reasoning. Claude's knowledge of food and geography is passable; its understanding of proverbs , the heart of Akan oral tradition, is poor in both languages and worse in Twi. We release all data and code publicly and hope this work encourages the community to treat cultural knowledge as a first-class evaluation criterion when assessing LLMs for deployment in African contexts.

Acknowledgements

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References

David I. Adelani et al. 2021. MasakhNER: Named entity recognition for African languages. *Transactions of the ACL*.

David I. Adelani et al. 2022. A few thousand translations go a long way! In *Proceedings of NAACL*.

David I. Adelani et al. 2024. IrokoBench: A new benchmark for African languages in the LLM era. In *Proceedings of EMNLP*.

Anthropic. 2024. The Claude 3 model family: Opus, sonnet, haiku. Technical report.

Paul Azunre et al. 2021. NLP for Ghanaian languages. In *Proceedings of the AfricaNLP Workshop, EACL*.

Yi Fung, Ruining Zhao, Jae Doo, Chenkai Sun, and Heng Ji. 2024. Massively multi-cultural knowledge acquisition & LM benchmarking. *arXiv:2402.09369*.

Dan Hendrycks et al. 2021. Measuring massive multitask language understanding. In *Proceedings of ICLR*.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in NLP. In *Proceedings of ACL*.

Shayne Longpre et al. 2023. The Flan collection: Designing data for effective instruction tuning. In *Proceedings of ICML*.

Meta AI. 2024. The Llama 3 herd of models. *arXiv:2407.21783*.

Shamsuddeen H. Muhammad et al. 2023. AfriSenti: A sentiment benchmark for African languages. In *Proceedings of EMNLP*.

Jessica Ojo et al. 2025. AfroBench: How good are LLMs on African languages? In *Findings of ACL*.

OpenAI. 2024. GPT-4o system card. Technical report.

Anastasios Romanou et al. 2024. INCLUDE: Evaluating multilingual language understanding with regional knowledge. *arXiv:2406.09948*.

Weiyan Shi et al. 2024. CulturalBench: Measuring cultural knowledge of LLMs. *arXiv:2410.02677*.

Chenxi Whitehouse, Faisal Hamman, and Anders Søgaard. 2023. LLM-powered data augmentation for enhanced cross-lingual performance. In *Proceedings of EMNLP*.

8 Dataset Schema

Each item in AfriCulture-QA is stored as a CSV row with seven fields:

`id`	unique integer
`topic`	domain tag (one of four)
`question_en`	question in English
`question_twi`	question in Twi
`answer_en`	answer in English
`answer_twi`	answer in Twi
`source`	source of data
`note`	additional information of the answers

The `topic` field uses one of: `food_and_cuisine`, `festivals_and_customs`, `proverbs_and_sayings`, `clothing_and_geography`.

9 Additional Dataset Examples

Domain	Question	Answer
Food	What is waakye?	Rice and beans cooked together, often with red sorghum leaves.
Festivals	Which festival honours Ashanti ancestors?	Awukudae or Akwasidae.
Proverbs	What does *'Obra ye kwan'* mean?	Life is a journey.
Clothing	What is Kente cloth associated with?	Ashanti and Ewe royalty and culture.

Table 3: Additional dataset examples (English only).